

STACK

Q3) Next Element With Smaller Digit-Sum

DS	Session	Stack	Question Information	Level 1	Challenge 43
Problem Description: You are given an array A of Q integers and Q queries. In each query, you are given an integer i ($1 \leq i \leq N$). Your task is to find the minimum index greater than i ($1 \leq i \leq N$) such that: 1. Sum of digits of A_i is greater than the sum of digits of A_j 2. $A_i < A_j$ If there is no answer, then print -1. Constraints $1 \leq N, Q \leq 10^5$ $1 \leq A_i \leq 10^9$ $1 \leq Q_i \leq N$ Input format • The first line contains two numbers N and Q. • The next line contains N numbers. • Next Q lines contain Q queries. Output format Print the answer as described in the problem					

CODE:

```

C main.c
1  #include <stdio.h>
2
3  int sum(long long x) {
4      int s = 0;
5      while (x) { s += x % 10; x /= 10; }
6      return s;
7  }
8
9  int main(void) {
10     int n, q;
11     if (scanf("%d %d", &n, &q) != 2) return 0;
12
13     long long a[n + 1];
14     int d[n + 1], ans[n + 1], s[n + 1], top = -1;
15
16     for (int i = 1; i <= n; i++) {
17         scanf("%lld", &a[i]);
18         d[i] = sum(a[i]);
19         ans[i] = -1;
20     }
21
22     for (int j = 1; j <= n; j++) {
23         int buf[n + 1], bTop = -1;
24         while (top >= 0) {
25             int i = s[top--];
26             if (a[j] > a[i] && d[j] < d[i]) ans[i] = j;
27             else buf[++bTop] = i;
28         }
29         while (bTop >= 0) s[++top] = buf[bTop--];
30         s[++top] = j;
31     }
32
33     int queries[q];
34     for (int k = 0; k < q; k++) {
35         scanf("%d", &queries[k]);
36     }
37
38     for (int k = 0; k < q; k++) {
39         printf("%d\n", ans[queries[k]]);
40     }
41
42     return 0;
43 }

```

TEST CASE 1 OUTPUT:

```

PS C:\Users\QH.SC.L\4\CSE25255\Projects\test> ./main
5 4
33 28 50 19 100
1
2
3
4
3
3
5
5

```

TEST CASE 2 OUTPUT:

```
13 cd /Users/danish@DESKTOP:~/Projects/leetcode/7/1021
```

```
6 3
```

```
73 45 61 88 120 50
```

```
1
```

```
2
```

```
3
```

```
4
```

```
3
```

```
5
```